

Megh Bahadur KC**Address:** Buffalo, NY, USA, 14214 | **Phone:** +1 (435) 881-2066**E-mail:** kc.megh2048@gmail.com, meghbaha@buffalo.edu
[visit my website](#) | [LinkedIn](#) | [GitHub](#) | [ResearchGate](#) | [Google Scholar](#)**PROFESSIONAL SUMMARY**

Ph.D. candidate in Civil Engineering at the University at Buffalo, specializing in sustainable public transport, freight systems and intelligent mobility through optimization, machine learning and deep reinforcement learning. Over 8 years of combined research, academic and industry experience developing efficient optimization tools, predictive models, techno-economic analyses, advanced data analytics and inclusive planning/policies at NREL, NYCDOT, Utah State University and the Department of Roads. Proven achievements include 7+ peer-reviewed publications, 17+ conference presentations, 2 internships, multiple awards and fellowships, effective undergraduate and graduate level teaching and hands-on project management as a Highway Engineer in transport infrastructure design and implementation.

EDUCATION

- **Ph.D.** in Transportation System Engineering; **University at Buffalo, SUNY**- GPA: 4.00, by May 2026
Dissertation Title: “**Accelerating Bus Transit Electrification through Optimization, metaheuristics and Reinforcement Learning Frameworks**” Advisor: Dr. Ziqi Song, Associate Professor
- **M. Sc.** in Construction Management; Pokhara University, Nepal- GPA: 3.95
- **B. E.** in Civil Engineering; Institute of Engineering, Tribhuvan University, Nepal; ranked top 4

RESEARCH INTERESTS

- **Operation Research:** Optimization, Application of operation research on electrified transportation systems, Reinforcement Learning for transport system efficiency, Multimodal Freight modeling
- AI/RL/ML in transportation, ITS and advanced transportation applications (V2X, CAV), Digital twins
- Travel Demand Modeling, Travel Behavior & Discrete Choice Modeling
- Inclusive Planning & Policy for urban land use & transportation system analysis

WORK EXPERIENCE

SN.	From	To	Employer	Position	Supervisor
7	01- 2025	to date	University at Buffalo, SUNY	Graduate Research Assistant (GRA)	Asso. Prof. Dr. Ziqi Song
6	09-2025	to date	New York City Department of Transportation	Freight Analytics Intern	Catherine Ponte, Team lead
5	05-2024	01-2025	National Renewable Energy Laboratory, Denver, CO	Transport System Analysis Intern	Jason Lustbador, Group Manager
4	08- 2021	01-2025	Utah State University	GRA	Asso. Prof. Dr. Ziqi Song
3	03- 2020	08- 2021	Pokhara University, Nepal	Asst. Professor	Dr. KN Pandey, Principal
2	07- 2015	08- 2021	Ministry of Physical Infrastructure and Transport, Nepal	Highway Engineer	Director General, Department of Roads
1	03- 2014	02- 2015	Tribhuvan University, Nepal	Assistant Lecturer	Prof. Dr. Rajan Suwal

1) Graduate Research Assistant @ University at Buffalo

- Ongoing PhD Dissertation research chapters:
 - *Multi-Agent Deep Reinforcement Learning for Integrated Bus Scheduling and Charging Management Under Uncertainty in Multi-Line Public Transit Operation.*
 - *Deep Reinforcement Learning Framework for Simultaneous Fleet Optimization and Charging for Multi-line Bus Transit System in a Dynamic Environment.*
 - *Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Bus System Including Demand Charges & Battery Degradation.*
 - *Bus Electrification: Fleet Optimization and Charging Management Considering Partial Recharges, TOU and Demand Charge.*

- Led "Residential Location Choice Modeling" project for WFRC Utah, identified key policy intervention for TOD & pedestrian-friendly infrastructure, contributing to an increase in transit ridership
- Developed and published freight-analytics-dashboard Python package on PyPI

Grant & Proposal Writing Contribution

- NYSERDA RFP 5965- *Electric School Bus Transition Support Contractor*, \$1M, 2025-2028
 - Drafted Statement of Work (SoW) section [submitted]
- DOE, Joint Office of Energy and Transportation (JOET), Concept paper in response to FOA – *Expanding E-Mobility Solutions through Electrified School Buses*, \$250,000.
 - Drafted Background and Proposed Methodology section

2) Freight Analytics Intern @ New York City Department of Transportation (NYCDOT)

- Designing a VMT assessment and GHG reduction tool for NYC freight transportation
- Quantifying the effect of freight policy intervention such as NY congestion pricing, off-hour delivery programs in terms of VMT, GHGs and air pollution
- Complex public and private dataset analysis including StreetLight, INRIX, Geotab, ATRI

3) Transport System Analysis Intern @ National Renewable Energy Laboratory (NREL)

- Developed & coded Freight Analysis Framework (FAF) & CFS flow containerization model & calibration
- Robust optimization of intermodal freight operations in disruptive events (national-scale) through modeling & simulation of containerized freight flow
- Built last-mile delivery vehicle routing: VRP, CVRP, VRPPD, and benchmarked with Google OR-Tools
- Contributed on SMART 2.0 Freight project by successfully developing & integrating the electric variant of NREL's open-sourced Freight Integrated Simulation Model (FRISM) for heterogeneous mixed fleet combination using VNS metaheuristics
- Formulated methodological framework for the county-level freight flow disaggregation model algorithm
- Created & hosted an interactive Container Freight data visualization Dashboard in Streamlit
- Received Excellent (best) performance grade during review for the year from the Group Manager

4) Graduate Research Assistant @ Utah State University

- Developed and linearized a multi-objective optimization model in GAMS and produced a hybrid metaheuristic algorithm in Python 3.10 with solver efficiency (730 seconds) in large-scale networks
- Produced and enhanced a multistage multi-objective linear /nonlinear optimization model with the achievement of efficient recharging and scheduling of EVs in 1,000+ node networks in 2 stages
- Automated a space-time prism (STP) modeling by developing an ArcGIS Pro algorithm in Python using the arcpy environment to measure accessibility for people with disabilities in only 1.5 months
- Performed techno-economic analysis on "Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Electric School Bus System Including Demand Charges & Battery Ageing."
- Led the best group project in machine learning titled "Fake News Detection: Investigating ChatGPT" by using Naive Bayes, Random Forest, and Neural Networks among 19 groups
- Conducted research on Discrete Mode Choice Modeling using Nested and Multinomial Logit in Pandas Biogeme, resulting in policy recommendations for the Inclusive Transportation System in Utah
- Completed term project on Change Detection Analysis for Selected Municipalities in Utah using ArcGIS Pro
- Conducted emission modeling for a new electrified metro line system, estimating mode shift, which reduced 6% annual GHG
- Participated in Kaggle Competition as a part of Final Project on Machine Learning and Data Analytics

5) Highway Engineer/ Site Project Manager

- Completed feasibility studies for multi-million-dollar Highway Projects, cost-benefit analysis, specifications.
- Obtained design approval for a 17-mile 6-lane road project, representing the Government of Nepal.
- Managed the procurement process for a massive ~\$40M road project, including bid document preparation, contract work specification, bid publication, evaluation, and contract award.

- Served as a site project manager, leading a team of 3 engineers, 8 sub-engineers, and 15 site supervisors.
- Skillfully coordinated with the communication manager for the upper body during the project's execution.
- Recognized with the Best Employee Award in 2019 for reducing office annual costs by 19% through optimizing employee schedules and operations.
- Part-time Assistant Professor, lectured to first-year MSc students on "Project Planning and Control."

RESEARCH EXPERIENCE

Peer-Reviewed Journal Articles

- **KC, M. B., & Song, Z.** (2026). School Transport Electrification - Adoption, Strategies, Methods and Policy: A Comprehensive Review. In arXiv e-prints. DOI: <https://doi.org/10.48550/arXiv.2601.00838>
- **KC, M. B., & Song, Z.** (2025). Evolving School Transportation: Integrated Dynamic Route Optimization and Partial Charging for Mixed Fleets. In arXiv e-prints. DOI: <https://doi.org/10.48550/arXiv.2511.00600>
- **KC, M. B.** (2025). Land Use Transition and City Desirability: Case Study of Cache County Over Two Decades. *Growth and Change, A Journal of Urban and Regional Policy*, 2025; 56(4). DOI: <https://doi.org/10.1111/grow.70057>
- **KC, M.B., Song, Z., Park, K. & Christensen, K.** (2025). *Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah*. In arXiv e-prints. DOI: <https://doi.org/10.48550/arXiv.2511.11748>
- **KC, M.B., Mishra, A.K. & Karki, R.** (2020). Perception Based Assessment of Bidding Practices and Effectiveness of E-Bidding System in Nepal. *East African Scholars of Economics, Business and Management* 2020; 3(7): 588-597. [Link](#)
- **Mishra, A.K., KC, M. B. & Aithal, P.S.** (2020). Association of Number of Bidders and Minimum Bid Ratio (AER) with Effect of E-bidding of Different Project. *International Journal of Management, Technology, and Social Sciences* 2020; 5(2): 201-215. [Link](#)
- **KC, M. B., & Mishra K.** (2019). Bidding Trends of Contracts based on Types and Sizes of Projects under Road Divisions Butwal and Shivapur. *Journal of Advanced Research in Construction and Urban Architecture* 2019; 4(4): 7-16. [Link](#)

Under Review and Under-Preparation Journal Papers

- **KC, M. B., & Song, Z.** (2026). Multi-Agent Deep Reinforcement Learning for Integrated Bus Scheduling and Charging Management Under Uncertainty in Multi-Line Public Transit Operation. *[work in progress]*.
- **KC, M. B., Song, Z. & Chaoyue Zhao.** (2026). Deep Reinforcement Learning Framework for Integrated Fleet and Charging Management for Electric Bus Transit System in Dynamic Environment. *[TRBAM 2026]*.
- **KC, M. B., & Song, Z.** (2026). Bus Electrification: Fleet Optimization and Charging Management Considering Partial Recharges, TOU and Demand Charge. *[TRBAM 2026]*.
- **KC, M. B., & Song, Z.** (2025). Integrated Optimized Charging Decision and Optimal Battery Sizing of V2G Enabled Electric School Bus System Including Demand Charges & Battery Ageing. *[TRB 2024, under review, Applied Energy]*.
- **KC, M. B., & Song, Z.** (2025). Multi-Stage, Multi-School Electric Bus Routing and Scheduling Optimization with recharging schedule and re-use of Buses using metaheuristics. *[TRB 2024 & under review, TR Part D]*.
- **KC, M. B., Song, Z., Park, K., Chamberlin, B. & Christensen, K.** (2025). Household Residential Location Choice Modeling for People with Disabilities in Utah. *[TRBAM 2026 & under review, TR Part A]*.

RESEARCH PROJECTS

- INtermodal Freight Optimization for a Resilient Mobility Energy System (**INFORMES**), Advanced Research Projects Agency-Energy's (ARPA-E), US **Department of Energy (DOE)** @ National Renewable Energy Laboratory, Denver, Co, USA
- Route Optimization, On-route Charging and V2G Integration on Transportation thrust for **ASPIRE** (Advancing Sustainability through Powered Infrastructure for Roadway Electrification) National Science Foundation (**NSF**) **Engineering Research Center**, Logan, Utah
- Increasing Affordability, Energy Efficiency, and Ridership of Transit Bus Systems through Large-Scale Electrification, US Department of Energy (DOE)

- Systems and Modeling for Accelerated Research in Transportation **SMART 2.0, Vehicle Technologies Office**, US Department of Energy (DOE) @ National Renewable Energy Laboratory, Denver CO
- Feasibility Analysis of Electric Roadways Through Localized Traffic, Cost, Adoption, and Environmental Impact Modeling, **ARPA-E, DOE**
- Investigating the Feasibility of Introducing Alternative Fuel Vehicles into Maintenance Fleet, Utah Department of Transportation (**UDOT**)
- Enhancement in 4-step Travel Demand Modeling considering individuals with disabilities incorporating physical community and environmental context, Funded by National Institute on Disability, Independent Living, and Rehabilitation Research (**NiDILRR**)

TEACHING EXPERIENCE

- **Teaching Assistant**, University at Buffalo; Utah State University, USA: 2021- present
 - Taught, provided feedback & graded students on assignments in class and in office hours
- **Assistant professor**, Pokhara University, Nepal: 2020-2021
 - Project planning & Control, Graduate Course – achieved 100% pass rate for MSc 1st semester
- **Lecturer**, Tribhuvan University, Nepal: 2013-2014
 - Transportation Engineering, Undergraduate Course – improved pass rates than in previous year
 - Supervised 2 Undergraduate Final projects on transportation planning and design

SKILLS

- **Programming and Optimization:** Python, R, GAMS, GurobiPy, CPLEX
- **Mapping and Visualization Software:** ArcGIS Pro, QGIS, ArcPy, matplotlib, seaborn
- **Machine/Deep Learning Frameworks:** SVM, SVD, DQN, Pytorch, HPC, PPO, MAPPO, MADDPG
- **Transportation Demand Modeling:** CUBE, Synchro, TransCAD, VISSIM, SmartRoad
- **Project management software:** Microsoft Project
- **Others:** Jupyter, Pandas, NumPy, scikit-learn, API web scraping, NLP, SQL

CERTIFICATIONS & LICENCES

- Engineer In Training (EIT), NCEES ID:2387261
- General Engineer, Civil, Nepal Engineering Council (NEC), since 2015

PROFESSIONAL AFFILIATIONS

- Member, TRB Standing Committee on Transit Operations (AP014), 08/2025-04/2028
- Member, TRB Standing Committee on Bus Transit Systems (AP050), 2025-2025
- Member, TRB Sub-committee on Freight Modeling (AT015(1)), 2023-2025
- Member, American Society of Civil Engineers (ASCE), 2025-2026
- Student member, Institute of Transportation Engineers (ITE), 2023-2025

LEADERSHIP ACTIVITIES

- Member, Tau Beta Pi, Utah Gamma Chapter, 2023-2025
- Member, Student Library Advisory Board, USU, 2023/24
- Utah State University Graduate Student Council (GSC), 2023/2024
- Vice-President, Civil Engineering Student Society-Nepal (CESS), 2012/2013, IOE Pulchowk, Nepal

AWARDS

- **Winner – Fellowship (\$35,000)**, September 11th Memorial Program for Regional Transportation Planning; New York Metropolitan Transportation Council (NYMTC), 2025
- **Winner** - Institute of Transportation Engineers (ITE) Student Research Competition, Utah, 2024
- **Winner** - Prakash S Lad Scholarship for Courage and Perseverance in the College of Engineering (ENGR), Utah State University, 2023 (**\$2000.00**)
- Best student term project on data science titled "*Fake News Detection: Investigating ChatGPT.*"
- **Dean list and University topper** in MSc Construction Management, Pokhara University, 2021

- **Outstanding MSc Student of the Year**, Pokhara University, 2019.

MEDIA MENTIONS

- Weitekamp, R. (2025, Oct). *Scholars Announced for NYMTC's 2025-26 9-11 Memorial Fellowship Program*. [Media link here](#)
- Maxwell, A. (2025, Sept 17). *CSEE student awarded September 11th Memorial fellowship for AI driven transportation solution for system efficiency*. [Medial link here](#)
- Dahle, S. (2024, Jan 23). *USU Engineering Scholar Showcases Transportation Research*. [Media link here](#)

REFERENCES

Ziqi Song, PhD Associate Professor Dept. of Civil, Str. and Env. Eng. University at Buffalo, SUNY Email: zqsong@buffalo.edu Phone: +1-716-645-3393	Keith Christensen, PhD Professor & Department Head LAEP Utah State University, Utah Email: keith.christensen@usu.edu Phone: +1-435-797-0507	Jason Lustbador Group Manager AVCI, CIMS National Laboratory of the Rockies Email: jason.lustbader@nrel.gov
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PEER-REVIEWED CONFERENCE PRESENTATION

17. **KC, M. B.**, Song, Z., Park, K., Chamberlin, B. & Christensen, K. (2025). Household Residential Location Choice Modeling for People with Disabilities in Utah. *ASCE International Conference on Transportation Development*, June 6-8, 2025, Arizona, USA (Accepted)
16. **KC, M. B.**, & Song, Z. (2024). Multi-Stage, Multi-School Electric Bus Routing and Scheduling Optimization Using Metaheuristics. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC
15. **KC, M. B.**, & Song, Z. (2024). Integrated Optimized Charging and Economic Analysis of V2G Enabled School Bus System. *Transportation Research Board (TRB) Annual Meeting, January 5-9, 2025*, Washington DC
14. **KC, Megh** and Song, Ziqi (2024), "Inclusive Transport Policy: Mode Choice Behavior of People with Disabilities in Utah," *2024 ITE Mountain District Annual Meeting, Big Sky, Montana*, 19-21 June 2024 (Accepted)
13. **KC, Megh**; Song, Ziqi; Park, Keunhyun and Keith Christinsen (2024), "Inclusive Transportation System and Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah", *6-Minute Showcase for the 2024 TRB Annual Meeting as part of the Young Member Council - Sustainability and Resilience*, Washington DC (awarded as recognized presentation)
12. **KC, Megh** and Song, Ziqi (2024), "Evolving School Transportation: A Comprehensive Approach to Bus Electrification with Dynamic Route Optimization and Partial Charging for Mixed Fleets", *Transportation Research Board (TRB) Annual Meeting, January 7-11, 2024*, Washington DC
11. **KC, Megh**; Song, Ziqi; Park, Keunhyun and Keith Christinsen (2024), "Understanding Mode Choice Behavior of People with Disabilities: A Case Study in Utah", *Transportation Research Board (TRB) Annual Meeting, January 7-11, 2024*, Washington DC
10. **KC, Megh** and Song, Ziqi (2023), "School Transportation Electrification: Simultaneous Route Optimization, Timetabling, and Partial Charging for Mixed Fleet," *2023 ITE Mountain District Annual Meeting, St. George, Utah*, 21-23 June 2023
9. **KC, Megh** and Song, Ziqi (2023), "Electric School Buses: Optimal Routing and Partial Charging," *USU Student Research Symposium 2023*, Logan, 12 April 2023
8. **KC, Megh** and Song, Ziqi (2023), "School Bus Electrification: Route Optimization, Timetabling, and Partial Charging," *College of Engineering Research Week, USU, Logan*, 4 April 2023.

RESEARCH OUTREACH ACTIVITIES

11. Reviewer: *Transportation Research Board Annual meeting, 2026*

10. Reviewer: *International Journal of Transportation Engineering and Technology (IJTET)*
9. Reviewer: *Journal of Transportation System and Engineering (JoTSE)*
8. Reviewer: *Energy 360*
7. Reviewer: *Transportation Research Record, Journal of Transportation Research Board*
6. Reviewer: *Transportation Research Board Annual meeting, 2025*
5. Reviewer: *Transportation Research Board Annual meeting, 2024*
4. Reviewer: *Transportation Research Board Annual meeting, 2023*
3. Reviewer: *Transportation Research Board Annual meeting, 2022*
2. Reviewer: *East African Scholars Journal of Economics, Business and Management*
1. Reviewer: *LEC Web-Based First International Conference on Engineering and Technology*